

SOFTWARE ENGINEERING LAB – 2 (UE24CS341A)

Agile Backlog Creation & Sprint Simulation in Jira

Team Number: 003

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Problem Statement 3: Campus Placement & Internship Pipeline

EPIC: Epic 4 – Offer Management & Data Security (FR-004 and NFR-001)

Note: This report covers only Epic 4, my individual contribution from the Lab 1 report. The remaining requirements are covered separately by my teammates (Epic 1 – FR-001: Mounik; Epic 2 – FR-002 and NFR-002: Kshitij; Epic 3 – FR-003 and FR-005: Mallikarjuna).

Lab Objective

This lab converts the requirements FR-004 and NFR-001 from the Lab 1 report into an Agile Epic and User Stories in Jira, estimates effort using the Fibonacci scale, prioritises the backlog, simulates two one-week sprints on a Scrum board, and tracks sprint progress using the Burndown Chart.

1. Epic (Derived from FR-004 and NFR-001)

The two requirements are grouped into one Epic: FR-004 defines the offer-letter outcome of the placement pipeline (generate and send an offer once a student is selected), and NFR-001 defines how the sensitive data behind that outcome, student academic records and offer documents, must be protected.

Epic Key	Epic Name	Mapped Requirements	Description
T0-1	Epic 4: Offer Management & Data Security	FR-004, NFR-001	Enable Placement Officers to issue digital offer letters to students once they are selected for a drive, and protect student academic records and offer documents at rest using AES-256 encryption.

2. User Stories for Epic 4

The Epic is broken into 8 User Stories using the format: As a [role], I want [goal], so that [benefit].

Key	Story	As a...	I want...	So that...	Covers
T0-2	4.1 Mark student as selected	Placement Officer	to mark a student as "Selected" for a company drive	the offer letter process starts without manual follow-up	FR-004
T0-3	4.2 Generate offer letter PDF	Placement Officer	the system to generate an offer letter PDF with the student's and company's details filled in	every selected student gets a correct, consistent offer	FR-004
T0-4	4.3 Email the offer letter	Student	my offer letter emailed to me soon after I am selected	I get my offer quickly without chasing the placement cell	FR-004

Key	Story	As a...	I want...	So that...	Covers
T0-5	4.4 Offer failure alerts	Placement Officer	to be notified when an offer fails to generate or send	no selected student is left without an offer	FR-004
T0-6	4.5 Encrypt academic records	Student	my academic records (CGPA, backlogs, marksheets) encrypted at rest with AES-256	my sensitive data cannot be read if the storage is accessed without permission	NFR-001
T0-7	4.6 Encrypt offer documents	Student	my offer letters encrypted at rest with AES-256	only I and authorised staff can open them	NFR-001
T0-8	4.7 Secure key management	System Admin	encryption keys stored separately from the data and rotated regularly	the data stays protected even if one part of the system is compromised	NFR-001
T0-9	4.8 Security and latency benchmark	Developer	to run benchmark tests under simulated peak load	we can confirm encryption meets the security and latency targets	NFR-001

Acceptance criteria were taken from Lab 1. For example, FR-004: an offer letter PDF is generated and emailed to the student within 10 minutes of selection; a selected student who receives no offer letter, or an incorrect one, is a failure. NFR-001: all student academic records and offer documents are encrypted at rest using AES-256, and benchmarking tests confirm the target latency and security standards under simulated peak load.

3. Jira Backlog – Epic, User Stories and Story Points

The screenshot shows Epic 4 with all eight User Stories, the priority icon and story points on each row, and the Epic panel reporting 8 work items and an estimate of 29 points.

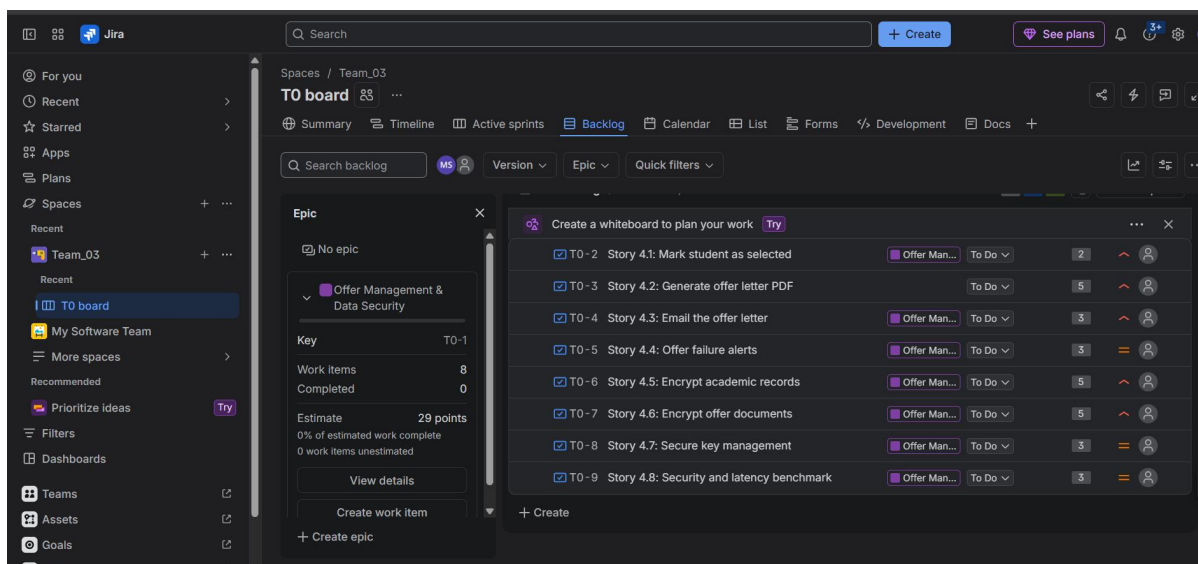


Figure 1: Jira Backlog – Epic 4 (T0-1) with stories T0-2 to T0-9 (2, 5, 3, 3, 5, 5, 3, 3 pts = 29 pts).

4. Backlog Prioritization

User Stories were ordered by the flow of the work and by dependency. The backlog order in Jira is 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, 4.8.

Key	Story	Priority	Story Points
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Key	Story	Priority	Story Points
T0-2	4.1 Mark student as selected	High	2
T0-3	4.2 Generate offer letter PDF	High	5
T0-4	4.3 Email the offer letter	High	3
T0-5	4.4 Offer failure alerts	Medium	3
T0-6	4.5 Encrypt academic records	High	5
T0-7	4.6 Encrypt offer documents	High	5
T0-8	4.7 Secure key management	Medium	3
T0-9	4.8 Security and latency benchmark	Medium	3

- **4.1, 4.2 and 4.3 (High):** these form the core offer flow of FR-004 (select, generate, email). Each step depends on the one before it, so they come first.
- **4.5 and 4.6 (High):** these encrypt the two data sets named in NFR-001, academic records and offer documents. They are High because they protect sensitive student data.
- **4.4 (Medium):** failure alerts cover the exceptional path of the offer flow. Placing it straight after the core flow means the first sprint delivers a complete and monitored offer pipeline.
- **4.7 and 4.8 (Medium):** key management and benchmarking support and verify the encryption, so they follow the encryption stories.

Total Story Points (Epic 4): 29 | High: 20 | Medium: 9

5. Story Point Estimation

Story points were assigned on the Fibonacci scale (1, 2, 3, 5, 8, 13...) relative to complexity, amount of work, and risk or uncertainty. The smallest story was sized 2, and the stories involving document generation and encryption of stored data were sized 5 because they carry the most uncertainty.

Key	Story	Points	Basis for the estimate
T0-2	4.1 Mark student as selected	2	Simple status change that triggers the offer process; low uncertainty.
T0-3	4.2 Generate offer letter PDF	5	Fills a template with student and company data and renders a PDF; incorrect details are a failure case, so accuracy needs care.
T0-4	4.3 Email the offer letter	3	Well-understood email integration, but must meet the 10-minute delivery target.
T0-5	4.4 Offer failure alerts	3	Detect generation or email failures, notify the officer, and allow a resend.
T0-6	4.5 Encrypt academic records	5	AES-256 at rest for stored records, while keeping access for authorised users working; existing data adds risk.
T0-7	4.6 Encrypt offer documents	5	Same encryption for stored offer files, plus a secure download path for the student.
T0-8	4.7 Secure key management	3	Keys stored apart from data and rotated without losing access to existing records.
T0-9	4.8 Security and latency benchmark	3	Run and record benchmark tests under simulated peak load.

6. Sprint Planning & Simulation

Sprint 1 (Duration: 1 week)

Sprint Goal: A selected student receives a correct offer letter by email, and the Placement Officer is alerted if delivery fails.

Key	Story	Priority	Pts	Status
T0-2	4.1 Mark student as selected	High	2	Done
T0-3	4.2 Generate offer letter PDF	High	5	Done
T0-4	4.3 Email the offer letter	High	3	Done
T0-5	4.4 Offer failure alerts	Medium	3	Done

Sprint 1 Velocity: 13 pts committed | 13 pts completed | 0 pts rolled over

Sprint 2 (Duration: 1 week)

Sprint Goal: Student academic records and offer documents are encrypted at rest with AES-256, keys are managed securely, and the result is benchmarked under simulated peak load.

Key	Story	Priority	Pts	Status
T0-6	4.5 Encrypt academic records	High	5	Done
T0-7	4.6 Encrypt offer documents	High	5	Done
T0-8	4.7 Secure key management	Medium	3	Done
T0-9	4.8 Security and latency benchmark	Medium	3	Done

Sprint 2 Velocity: 16 pts committed | 16 pts completed | 0 pts rolled over

Jira Sprint Board (Active Sprint view)

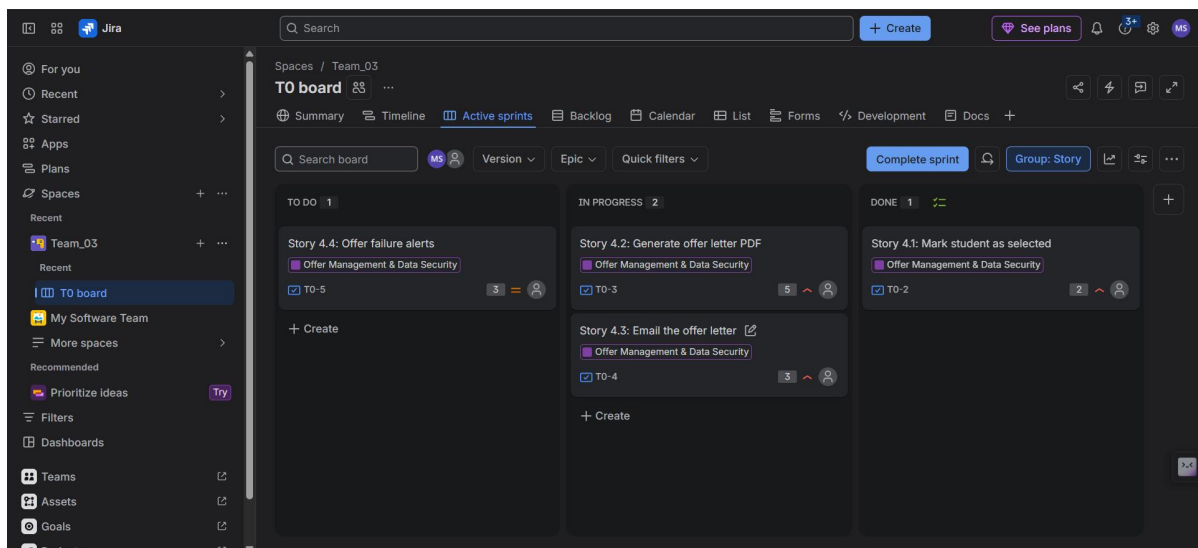


Figure 2: Sprint 1 Active Sprint board mid-sprint – Story 4.1 in DONE; Stories 4.2 and 4.3 IN PROGRESS; Story 4.4 in TO DO.

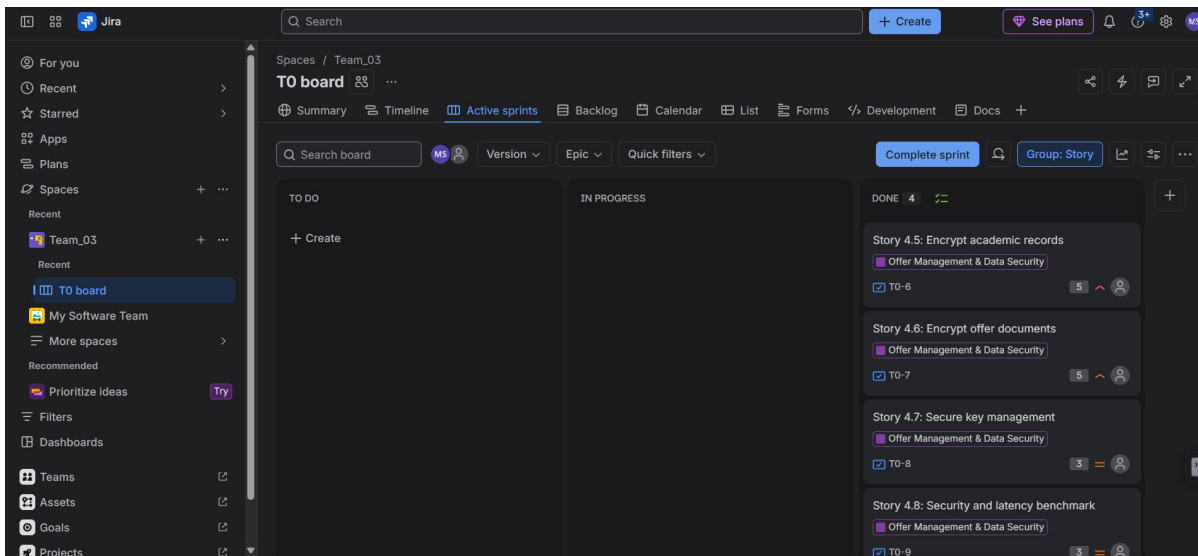


Figure 3: Sprint 2 Active Sprint board – all four stories (16 pts) in DONE, before clicking Complete sprint.

Sprint 2 outcome

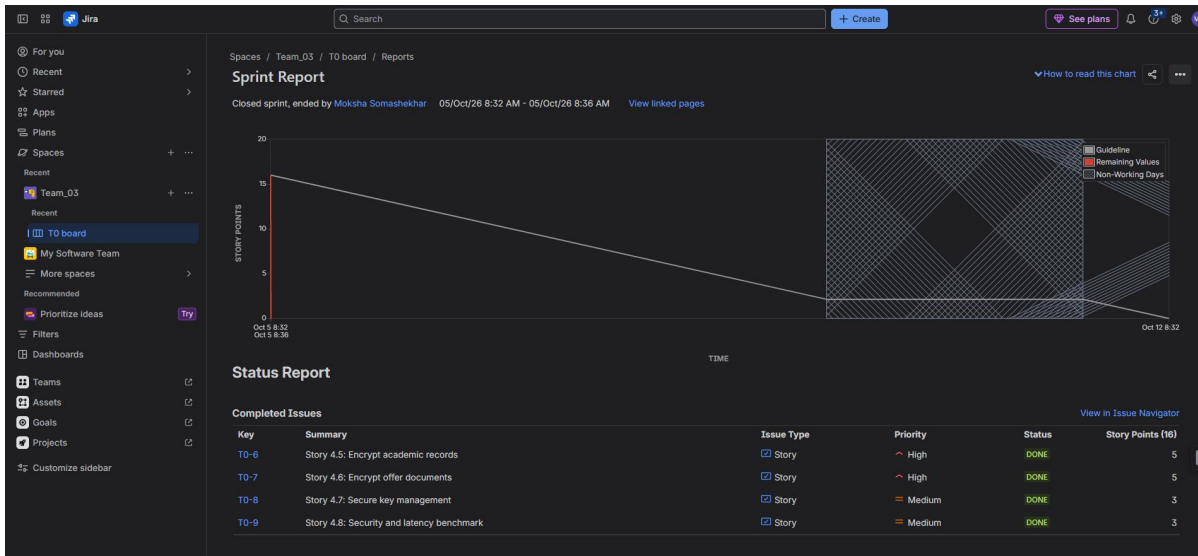


Figure 4: Sprint 2 outcome (Sprint Report, taken after completion) – T0-6 to T0-9 DONE, 16 story points.

7. Burndown Chart Analysis

The Jira Burndown Chart plots story points remaining (red line) against the ideal guideline (grey line). Each sprint is shown twice: once with non-working days displayed (hatched) and once with them hidden, which makes the guideline easier to read. The event table under each chart lists the exact completion times.

Sprint 1 Burndown (13 pts)

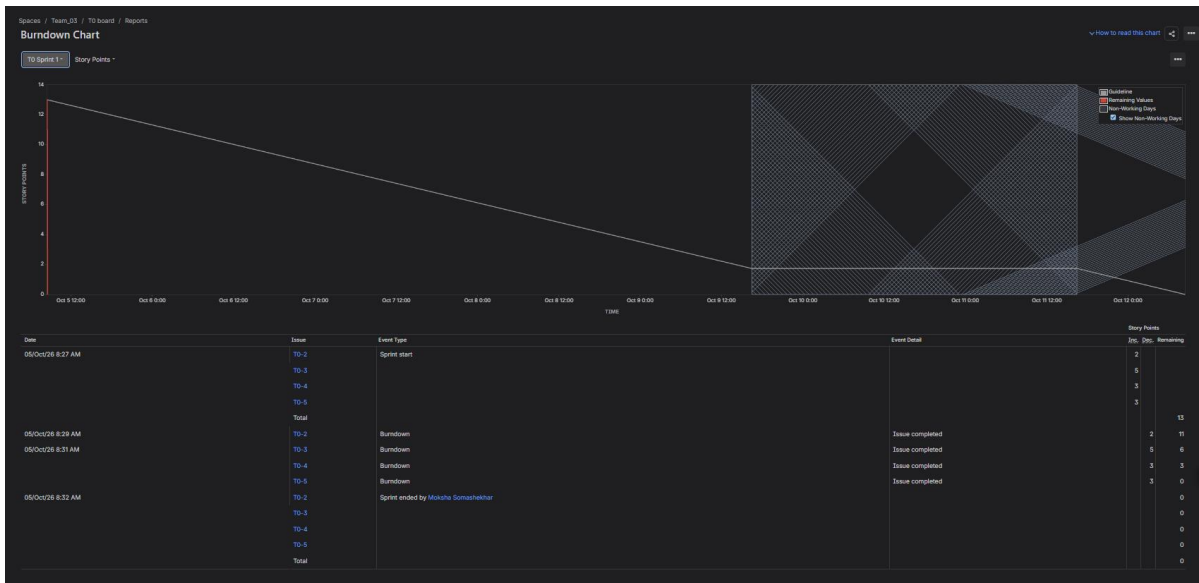


Figure 5: Sprint 1 Burndown, non-working days shown. 13 → 11 → 6 → 3 → 0 between 8:29 AM and 8:31 AM on 5 Oct.

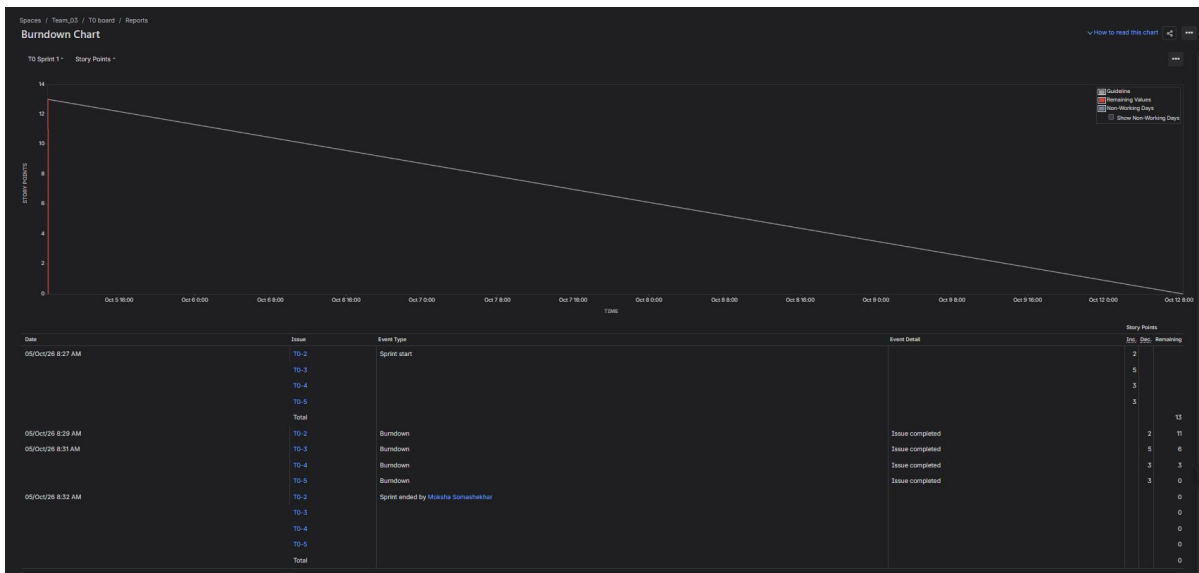


Figure 6: Sprint 1 Burndown, non-working days hidden.

Sprint 2 Burndown (16 pts)



Figure 7: Sprint 2 Burndown, non-working days shown. $16 \rightarrow 11 \rightarrow 6 \rightarrow 3 \rightarrow 0$ between 8:33 AM and 8:36 AM on 5 Oct.

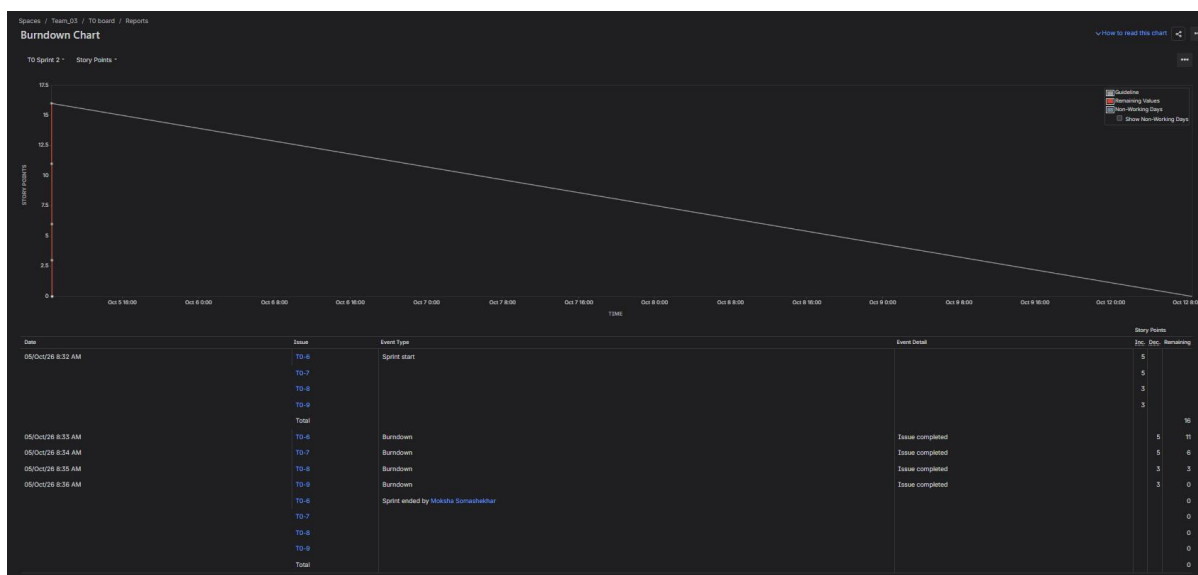


Figure 8: Sprint 2 Burndown, non-working days hidden.

Summary of the burndown data (from the Jira event tables)

Sprint	Start (points)	Completion events	End (points)
Sprint 1	8:27 AM – 13	8:29 AM T0-2 (-2) → 11; 8:31 AM T0-3 (-5) → 6, T0-4 (-3) → 3, T0-5 (-3) → 0	0
Sprint 2	8:32 AM – 16	8:33 AM T0-6 (-5) → 11; 8:34 AM T0-7 (-5) → 6; 8:35 AM T0-8 (-3) → 3; 8:36 AM T0-9 (-3) → 0	0

8. Reflection Questions

1. Did your estimations reflect the actual effort?

We can only partly judge this. The sprints were simulated by moving cards on the board, so no real development effort was spent and the estimates cannot be checked against actual hours.

What the exercise did show is that the relative sizing was sensible: the stories that involve generating the offer PDF (4.2) and encrypting stored data (4.5 and 4.6) received the highest estimate (5) because they carry the most uncertainty, while marking a student as selected (4.1) was sized 2 as the simplest change. In a real sprint I would expect 4.5 and 4.6 to be the most likely to overrun, since encrypting data that already exists while keeping access working for authorised users usually hides extra work. I would also watch 4.7: key rotation without losing access to existing records may turn out larger than 3 points. No story needed an estimate above 5, so none had to be split.

2. Was your backlog well-prioritized?

Mostly yes. The core offer flow (4.1, 4.2 and 4.3) was placed first, followed by failure alerts (4.4), so the first sprint ended with a complete, monitored path from selection to an emailed offer. Encryption, key management and benchmarking came in the second sprint, since the benchmark can only be meaningful once the encryption exists. One weakness is that the backlog was ordered by delivery flow rather than strictly by priority label: the Medium story 4.4 sits ahead of the High stories 4.5 and 4.6. A bigger concern is that offer letters are generated in Sprint 1 but only encrypted in Sprint 2, so for one sprint they would be stored unprotected. In a real project I would move 4.6 earlier, or build encryption in from the start.

3. How did your simulated sprint align with your plan?

Both sprints were completed as planned. Sprint 1 committed and completed 13 points and Sprint 2 committed and completed 16 points, a total of 29 points with nothing rolled over, and both sprint goals were met. In Sprint 1, story 4.1 was completed first and the other three stories were completed together shortly before the sprint was closed; the mid-sprint board shows cards in all three columns. Sprint 2 stories were completed one at a time, in order. Because the work was simulated, each sprint was finished within minutes (8:27 to 8:32 AM and 8:32 to 8:36 AM) instead of the one week that was planned, so the 100% completion reflects the simulation, not real delivery capacity.

4. What insights did the burndown chart give about your team's capacity?

In both sprints the red line dropped to zero within minutes of the start, far below the grey guideline, which is why it appears as a thin line at the left edge of the chart. In a real sprint that pattern would indicate that the team had under-committed. The event tables still gave useful information: Sprint 1 burned down 13 points and Sprint 2 burned down 16 points, so a realistic velocity would be somewhere around 13 to 16 points per week (about 14.5 on average), but this would need to be re-measured with real work. The two charts also differ in shape. Sprint 2 burned down evenly, one story per step (16 → 11 → 6 → 3 → 0), whereas in Sprint 1 three stories closed at the same time, which hides progress and would look like work being completed in one batch at the end. Keeping stories small, with none above 5 points, helped keep each step of the burndown modest.